New trading relationships can change the dominant vehicle.

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Evidence from DeFi

Jiahua Xu

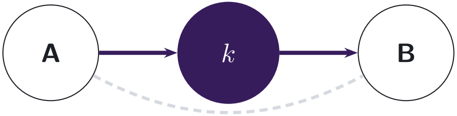
A cross-border payment needs someone to make the FX market

THE LIQUIDITY-PROVIDER PROBLEM

A payment provider receives one currency but must deliver another it does not hold.

more than 85%

of FX transactions involve the U.S. dollar



Currency k bridges an illiquid pair.

vehicle currency

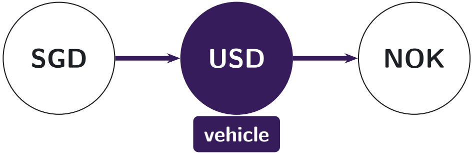
Winning the middle concentrates trading and liquidity in the vehicle.

Pool routes reveal the vehicle currency

SINGAPORE PAYMENT ANALOGY

A Singapore-dollar payment to Norway may be converted through US dollars.

Conventional turnover data record two legs. Pool logs preserve their connected route.



One instruction;
two linked markets.

DeFi is on-chain finance executed by smart contracts. Decentralised exchanges trade tokens through liquidity pools; **stablecoins** target a reference currency, usually the U.S. dollar.

One transaction reveals the connected route

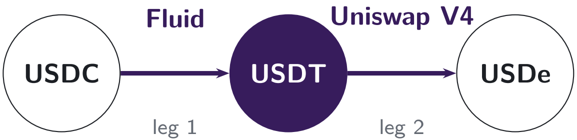
ROUTE PANEL

475 million
pool-level swaps

November 2018–June 2026

9
DEX deployments

OBSERVED POOL ROUTE



Pair: USDC $\rightarrow$ USDe Vehicle: USDT

The ordered sequence of legs is the route.

V2 turns a mandate into a market choice

V1: NOVEMBER 2018

217,003

of token-to-token routes pass through ETH

$\Longrightarrow$

V2: MAY 2020

95.5%

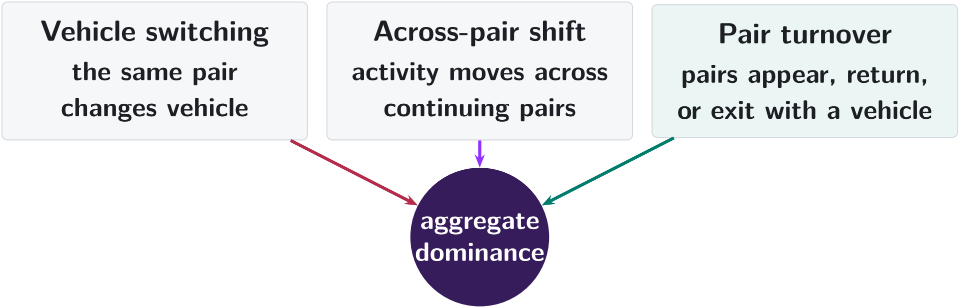
of single-leg trades use a WETH pool in 2026

97.9%

of new token combinations include WETH

The rule disappears; inherited liquidity can remain.

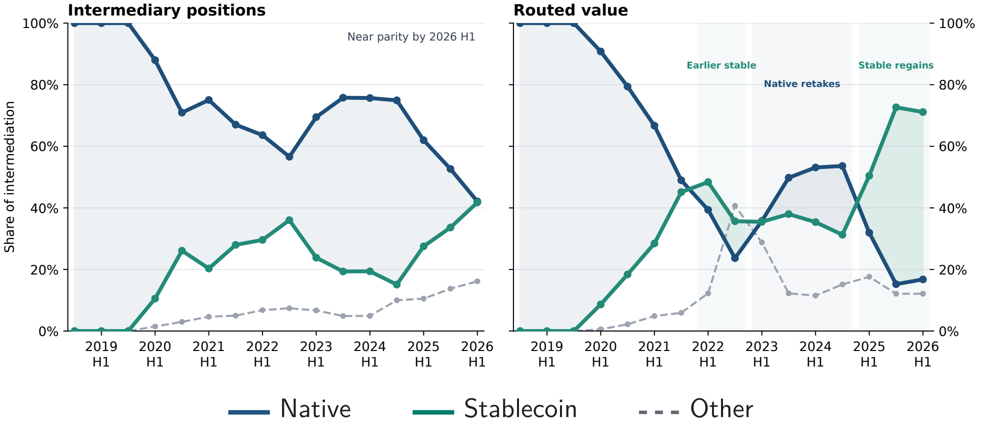
Aggregate dominance can change in three ways



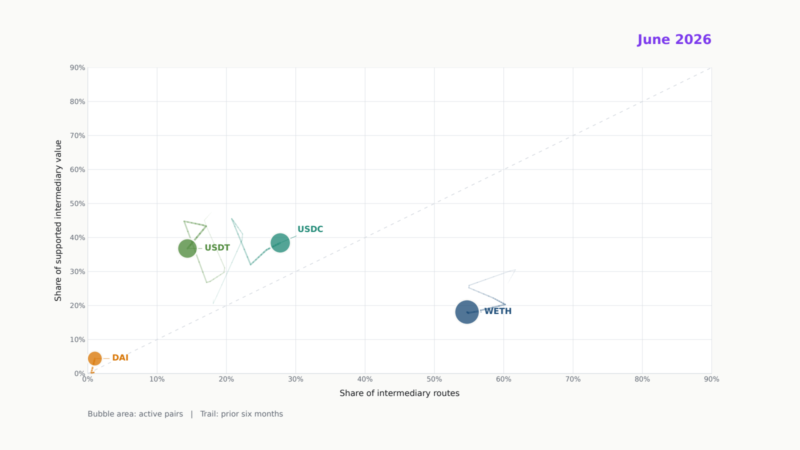
The decomposition locates the rotation; later evidence explains persistence.

Stablecoins regain the routed-value lead by 2026 H1

ALL ROUTE LENGTHS



Vehicle leadership turns over through time



▶ Play 18-second film

Pair formation supplies most of the rotation

EXACT DECOMPOSITION IN PERCENTAGE POINTS (PP): 2024 H1 TO 2026 H1

$$\Delta S = \underbrace{\bar{W}\sum_p \bar{\omega}_p\,\Delta s_p}_{\substack{\text{switching} \\ -0.1\ \textbf{pp}}} + \underbrace{\bar{W}\sum_p \bar{s}_p\,\Delta\omega_p}_{\substack{\text{reallocation} \\ +8.4\ \textbf{pp}}} + \underbrace{(\bar{S}_C - \bar{S}_E)\,\Delta W}_{\substack{\text{continuing weight} \\ -0.5\ \textbf{pp}}} + \underbrace{(1-\bar{W})\,\Delta S_E}_{\substack{\text{period-specific pairs} \\ +17.8\ \textbf{pp}}}$$

STABLECOIN SHARE **16.9%** $\longrightarrow$ **42.4%**

First-observed pair entry supplies $+20.1$ **pp of** $+25.5$ **pp; switching gives** -0.1 **pp.**

A pair carries its first vehicle into later trading

STABLECOIN SHARE AFTER ENTRY, PER 10 PP MORE STABLECOIN USE ON THE ENTRY DAY

	Days 1–30	Days 31–120
All retrading pairs	**+8.92 pp**	**+8.40 pp**
	(0.15 pp)	(0.18 pp)
At least 5 entry routes	+9.70 pp	+9.31 pp
	(0.19 pp)	(0.44 pp)
At least 10 entry routes	+9.65 pp	+9.35 pp
	(0.32 pp)	(0.60 pp)

Near one-for-one persistence, however precisely entry is measured.

Two-leg capital opens the stablecoin contest

THE EVENT DATE USES PRIOR-DAY CAPITAL ONLY

BRIDGE BECOMES VIABLE
$10k on each leg
after earlier WETH use

→

ADOPT WITHIN 30 DAYS
38.3%

→

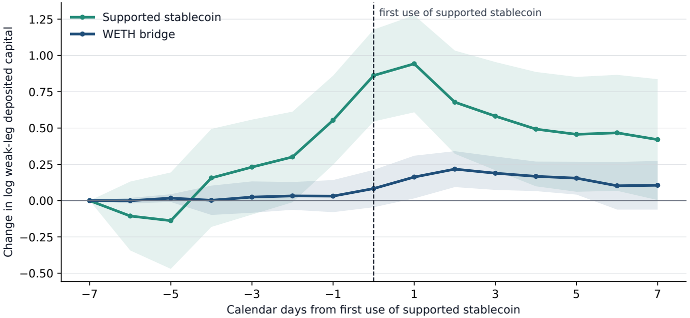
ADOPT WITHIN 120 DAYS
47.0%

Among first-month adopters, **62.4%** use a supported stablecoin again in days 30–119;
stablecoins carry **8.2%** of their routes in that window.

Viability comes first. Dominance develops more slowly.

Capital arrives before the first stablecoin route

SUPPORTED STABLECOIN FIRST USED AT DAY ZERO



Capital builds into first use, then partly unwinds.

BEFORE FIRST USE

$+0.86$

stablecoin bridge, day -7 to 0

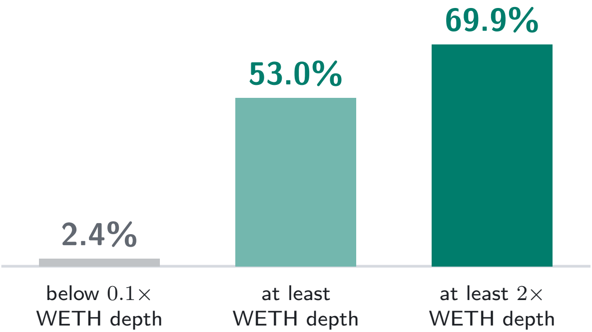
AFTER FIRST USE

-0.44

stablecoin bridge, day 0 to $+7$

Relative depth divides trading after the bridge forms

STABLECOIN ROUTE SHARE IN THE FIRST 30 DAYS AFTER PERSISTENT SUPPORT



$+10$ PP STABLE SHARE OF WEAK-LEG DEPTH

$+6.90$ **pp**
route share, days 0–29

$+8.35$ **pp**
route share, days 30–119

Vehicle competition turns on relative depth, leg by leg.

Current prices can overturn incumbency

STABLECOIN AND WETH PATHS BOTH FEASIBLE: SAME PAIR, INPUT, PRETRADE STATE

ENTRY FAMILY RETAINED

83.1%

of routes at the first feasible contest

INCUMBENT RETURNS MORE OUTPUT

93.3%

retain the incumbent

CHALLENGER RETURNS MORE OUTPUT

27.2%

retain the incumbent

Pair and date fixed effects: challenger price leadership changes incumbent retention by -58.08 **pp**.

Incumbency holds until the price lead moves.

When the price lead flips, route share follows

WITHIN-PAIR EXACT-PRICE CROSSING

MONTH -1

INCUMBENT LEADS

66.9%

incumbent share

price lead changes hands →

CROSSING MONTH

CHALLENGER LEADS

38.1%

incumbent share

ADDITIONAL FALL AT THE PRICE CROSSING

-29.0 pp

CHALLENGER PRIOR WEAK-LEG CAPITAL SHARE (+10 PP)

3.70 pp

probability the price lead lasts one month

Prices move flow. Earlier depth predicts which lead lasts.

Shortfalls cluster in young pairs and small trades

EXACT STABLECOIN-VERSUS-WETH COMPARISON; INPUT-VALUE WEIGHTS

ALL INPUT VALUES: GROSS → NET OF GAS

7.46 → 7.65 bp

nearly unchanged

INPUTS $100–999: GROSS → NET OF GAS

16.02 → 22.01 bp

fixed toll matters

PAIR AGE: 0–89 DAYS	PAIR AGE: 90–364 DAYS	PAIR AGE: 365+ DAYS
16.8 bp	**19.0 bp**	**2.1 bp**

Key takeaways

THE MAKING OF A DOMINANT VEHICLE CURRENCY

New pairs drive the rise	Entry vehicle persists	Depth divides the flow	Prices can flip it	Costs fade with age
$+20.1$ **pp**	$+8.92$ **pp**	**53.0%**	-58.08 **pp**	**2.1 bp**
of $+25.5$ pp	per 10 pp entry	at equal depth	retention swing	mature pairs
▸ 8	▸ 9	▸ 12	▸ 13	▸ 15

Two stages of currency competition

Which vehicle a new pair builds around; which survives later price and depth changes.

Implications for market design and oversight

Payment rails Nexus, Rialto, mBridge: access rules pick the currency that new corridors inherit. ▸ 8 ▸ 9

Stablecoin oversight Watch the bridge role, beyond issuance size: most routed value now crosses a stablecoin in the middle. ▸ 6

Challenger instruments Cheaper quotes in existing corridors dethrone nobody; challengers win by being liquid at formation. ▸ 8 ▸ 12

Liquidity at formation beats price competition later.

Implications for sovereign monetary policy

THE POLICY WINDOW OPENS WHEN NEW CORRIDORS FORM

BEFORE: MONITOR

Track which vehicle new corridors choose; aggregate dominance follows entry.

▸ 8

AT FORMATION: ACT

Be the liquid bridge on both legs at birth; the entry choice persists.

▸ 9 ▸ 12

AFTER: LOCKED IN

Late price cuts move flow only where challenger depth already exists.

▸ 8 ▸ 14

Design rules leave liquidity legacies: the V1 mandate outlived itself.

▸ 4

What we are asking next

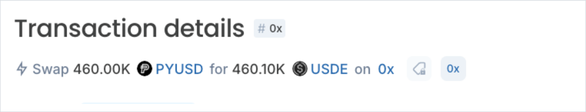
- Who supplies the bridge? Provider networks across pairs; issuer circulation into bridge capital; fee and incentive events.
- Does settlement architecture move liquidity? Uniswap V4 flash accounting against V3.
- Who earns the rents of dominance: traders, providers, or issuers?
- The same entry-first account of dominance, taken to conventional FX.

Comments on any of these are very welcome.

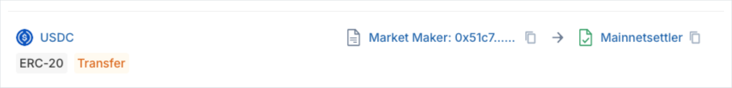
Appendix map

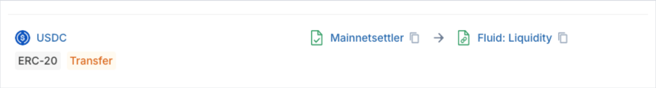
Pool data may start after the user instruction

USER INSTRUCTION RECORDED BY THE EXPLORER



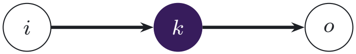
THE DECISIVE SETTLEMENT TRANSFERS





The maker supplies the USDC used by the Fluid leg. The instruction is PYUSD $\rightarrow$ USDe.

A1. One reconstructed route is one unit

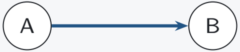


one route unit
two swap legs

- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

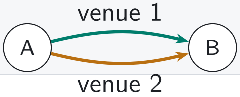
How we reconstruct a route from one transaction

Direct swap
one transaction

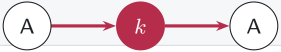


Sequential vehicle route
one transaction

Parallel split
one transaction

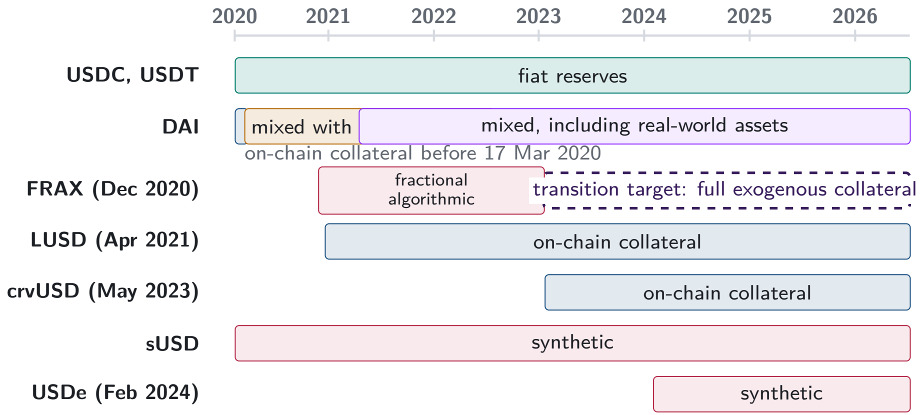


Round trip
one transaction



Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level routes reveal that choice inside aggregate turnover.

A2. Stablecoin backing changes over time



Peg parity preserves currency identity

Bahamian dollar $\longleftrightarrow$ **U.S. dollar**

one-for-one peg

Separate currency, monetary arrangements, and exchange controls

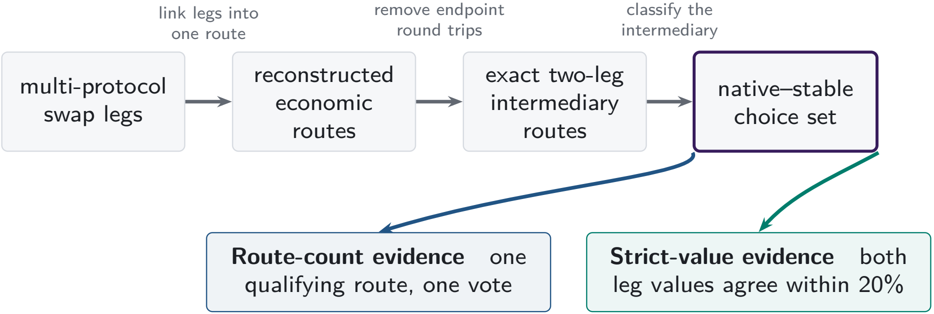
USDC $\longleftrightarrow$ **USDT** $\longleftrightarrow$ **DAI**

dollar target

Separate issuers, redemption, peg risk, and pool liquidity

Lower divergence loss may attract liquidity providers; issuers or affiliated market makers may also seed conversion pools. Peg stress reverses the comparison.

A3. One route universe supports two measurements



A4. State is reconstructed immediately before execution

1 verified starting state → **2** apply events in blockchain order → **3** recover state before the trade → **4** quote both routes

- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hook-dependent pricing
Constant product	■	–	–	–	–	■	–
Concentrated liquidity	■	■	–	–	–	■	–
StableSwap families	■	–	■	–	■	■	–
Weighted pools	■	–	–	■	■	■	–
V4 singleton	■	■	–	–	–	■	□

Executable-quote validation

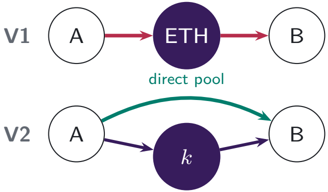
■ recovered pricing input
□ hook-dependent pricing unavailable

Protocol mechanisms studied

V4 singleton flash accounting and netting remain in scope; A5.3 shows the settlement mechanism.

A5.1. Pool formation determines available paths

V1 PROTOCOL RULE $\longrightarrow$ V2 MARKET CHOICE



V1 admits only ETH–token pools. V2 permits any ERC-20 pool, so market creation determines whether direct and vehicle paths coexist.

Which paths exist?

Economic implication: pool creation changes the feasible route set.

A5.2. Executable depth is path-specific

V3 CONCENTRATED-LIQUIDITY POSITIONS

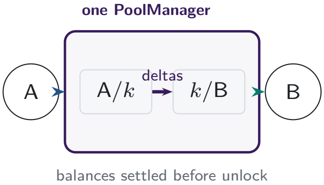
LPs choose a price range and fee tier for each pool. Positions covering the current price determine active liquidity in the direct pool and each vehicle spoke.

Which path offers depth at this trade size?

Economic implication: executable depth is path- and size-specific.

A5.3. Shared accounting moves route settlement

V4 SINGLETON AND LOCK ACCOUNTING



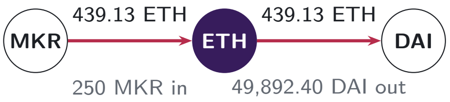
One PoolManager holds all pools. During a lock, the two route legs update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication: shared accounting changes where route obligations settle.

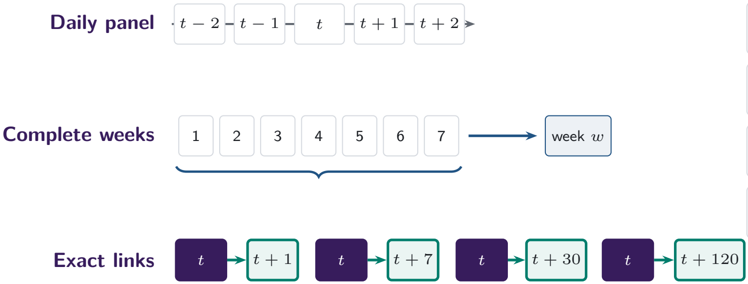
A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

A7. Daily and weekly frequencies answer different questions



Model role

Daily panel date fixed effects

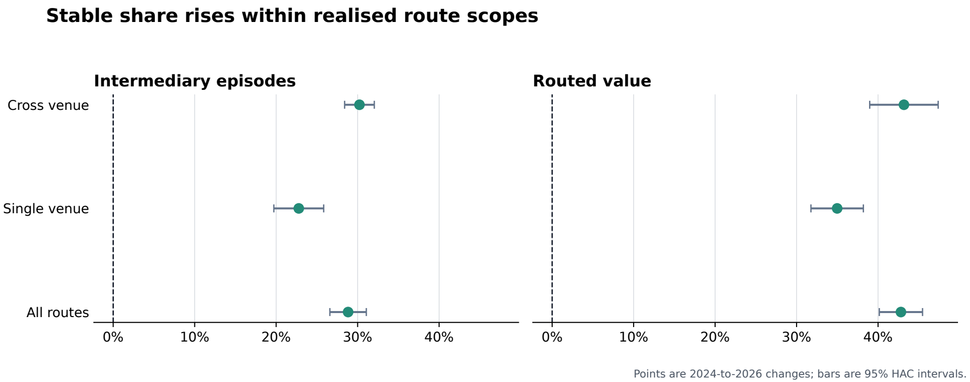
Calendar check week and weekday fixed effects

Complete weeks week fixed effects

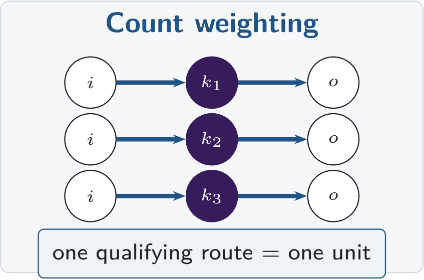
Exact links inference by origin date

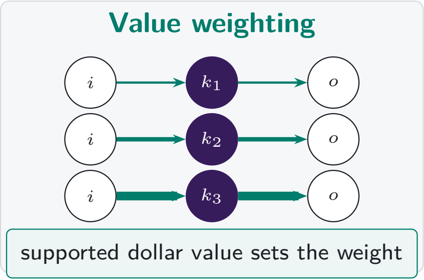
Exact horizons use the required target calendar date.

A7b. Stablecoin share rises within each venue scope



A8. Count versus value weighting





Vehicle dominance aggregates realised intermediary choices

One realised route

k occupies the intermediary position

Vehicle dominance

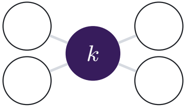
D_k = use through k
$\div$ total intermediary use

Value-weighted routes use comparable dollar amounts

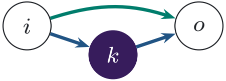
Excess vehicle use

$X_k = D_k \div E_k$
E_k: endpoint share of k

$X_k > 1$: use exceeds the endpoint benchmark



Network position



Execution cost

Endpoint use is the benchmark; network position and same-state cost remain separate.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I^{N}_{k,t} = \frac{n^{I}_{k,t}}{\sum_j n^{I}_{j,t}}$$

How often asset k sits inside a route.

Endpoint share

$$E^{N}_{k,t} = \frac{n^{E}_{k,t}}{\sum_j n^{E}_{j,t}}$$

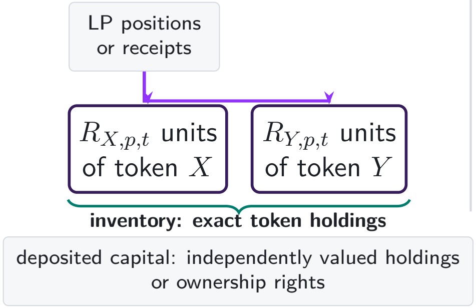
How often k appears at either observed route endpoint.

Observed pool-route endpoints supply the benchmark.

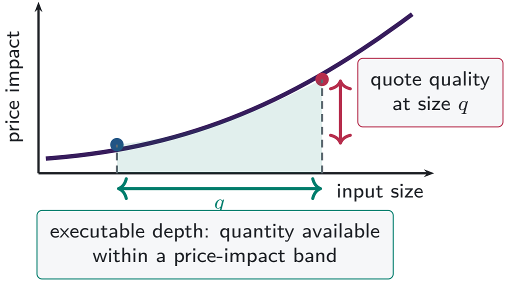
Excess use compares intermediation with observed route-endpoint use on the same support.

A10. Capital, inventory, and depth differ

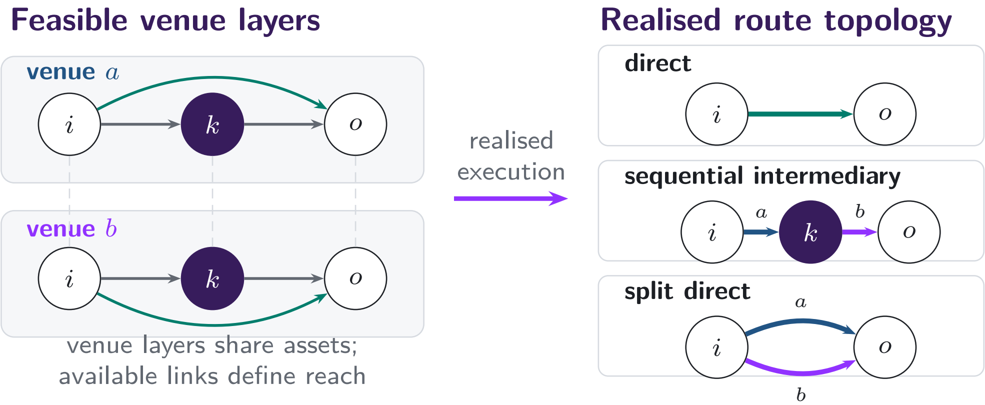
Pool state at time t



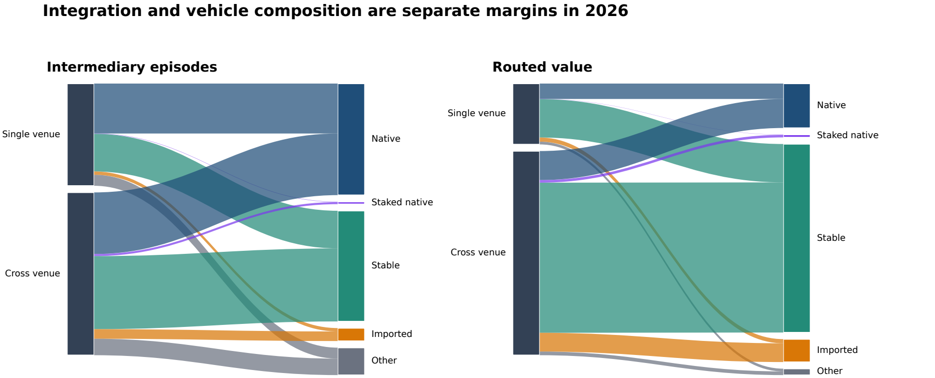
Execution from that state



A11. Additional venues expand the feasible route set



A11b. Venue scope and vehicle type in 2026



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

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- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Adams et al. (2024), *Uniswap v4 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Milionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage

A14. WETH stays the graph hub as realised use shifts

ALL ROUTE LENGTHS; MONTHLY TRADING GRAPHS

WETH BETWENNESS

#1

in all 8 annual graphs

0.925

in 2026 H1

WETH REALISED SHARE

76.2%

2024

42.4%

2026 H1

USDC + USDT REALISED SHARE

14.9%

2024

37.2%

2026 H1

Binary network position changes much less than actual intermediary use.

A15. Endpoint demand predicts intermediary use

	Intermediary share $(I_{a,t})$ [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	$+34.56$	$+34.55$	$+0.05$	—
	(23.04)	(23.04)	(0.56)	—
Stablecoin	$+2.42^{*}$	$+2.42^{*}$	$+0.85^{*}$	—
	(1.32)	(1.32)	(0.50)	—
Endpoint-demand share $(D_{a,t})$	—	—	$+1.59^{***}$	$+0.98^{***}$
	—	—	(0.03)	(0.30)
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

After date and currency effects, endpoint demand passes through almost one for one into intermediary use.

A16. Pair composition contains a mechanical component

WETH

WETH is an endpoint

↓

Eligible intermediaries: stablecoins

$+21.1$ **pp**

All matched pairs

WETH-linked route-count mass: 26.6% $\rightarrow$ 46.8%

$+3.7$ **pp**

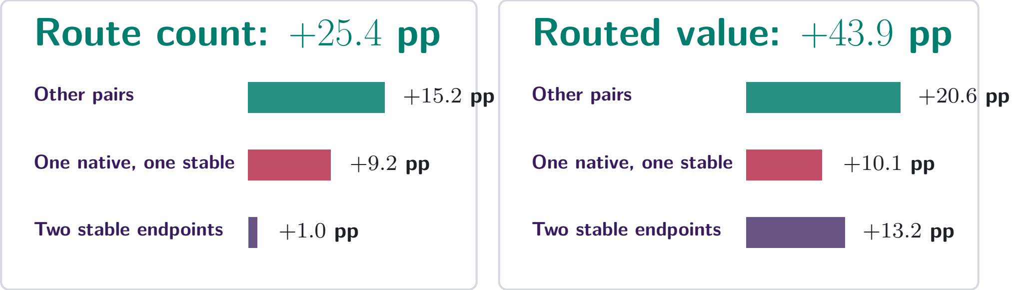
Pairs without WETH endpoints

79,347 matched groups

Changes within fixed pairs are $+0.2$ pp overall and $+0.3$ pp after removing WETH endpoints.

A17. Endpoint direction splits count and value

CONTRIBUTION TO THE STABLECOIN-SHARE INCREASE



USDT supplies $+13.7$ pp of the $+13.2$ pp two-stable-endpoint value channel.

A17b. Rotation survives time and endpoint restrictions

-0.4 pp to $+1.0$ pp
within-pair term across all adjacent H1 comparisons

1.1% $\rightarrow$ 9.0%
route-count share with nonvehicle endpoints

0.9% $\rightarrow$ 39.1%
supported-value share with nonvehicle endpoints

Pair entry and exit contribute $+7.2$ pp by count and $+33.8$ pp by value.

A18. Local depth remains informative alongside network reach

84.1%
of routes use the deepest supported bridge

$+6.79$ **pp**
route share per log point of weak-leg capital

$+9.10$ **pp**
route share per log point of same-day network reach

Local two-leg depth and broader network position both matter.

A19. First-use capital lies in active pools

SUPPORTED STABLECOIN, DAY -7 TO FIRST USE

92.5%
of first-use capital lies in continuing pools

$+0.16$
continuing-pool growth relative to WETH

$+6.9$ **pp**
greater probability of a newly active pool

A20. Relative depth predicts route flow

FIRST 30 DAYS AFTER PERSISTENT STABLECOIN SUPPORT

Below $0.1\times$ WETH depth	At least WETH depth	At least $2\times$ WETH depth
2.4%	53.0%	69.9%

Vehicle competition turns on relative depth.

A21. Usable depth turns support into first use

PROBABILITY OF FIRST STABLECOIN USE WITHIN 30 DAYS

BELOW $0.1\times$ WETH DEPTH

42.6%

$\rightarrow$

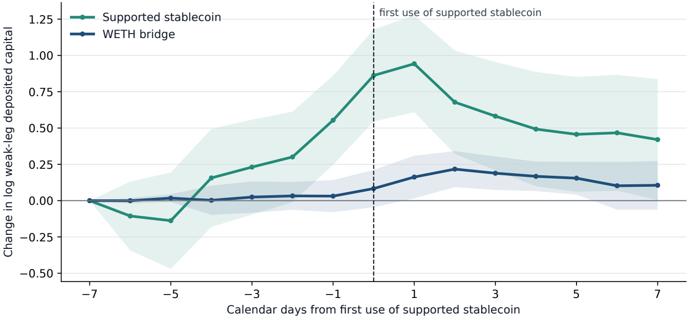
AT LEAST $0.1\times$ WETH DEPTH

84.1%

Positive pool capital becomes useful as both route legs deepen.

A22. Bridge capital builds before first use

SUPPORTED STABLECOIN FIRST USED AT DAY ZERO



Capital builds into first use, then partly unwinds.

BEFORE FIRST USE

$+0.86$

stablecoin bridge, day -7 to 0

AFTER FIRST USE

-0.44

stablecoin bridge, day 0 to $+7$

A23. Broader price search mostly changes the venue

SHARE OF ROUTES WITH A QUOTE MORE THAN 1 BP BETTER THAN EXECUTION

SAME VEHICLE, USED VENUES	→	SAME VEHICLE, ALL EXACT VENUES	→	ANY VEHICLE OR DIRECT
6.6%		**44.5%**		**46.4%**

Opening other vehicles adds 2.0 pp; stablecoin share changes by -1.2 pp.

A24. Persistence is equally strong on busy entry days

EFFECT OF 10 PP MORE STABLECOIN USE ON THE ENTRY DAY

	Days 1–30	Days 31–120
At least 5 entry routes	$+9.70$ **pp**	$+9.31$ **pp**
At least 10 entry routes	$+9.65$ **pp**	$+9.35$ **pp**

Stronger entry measurement leaves the persistence estimate intact.

A25. Divergence risk helps locate bridge capital

EFFECT OF 10 PP MORE PRIOR RELATIVE VOLATILITY

LOG BRIDGE CAPITAL

-0.117

current date (0.049)

-0.093

30 days later (0.029)

MEDIAN RELATIVE VOLATILITY

Native 126.9%

Stablecoin 148.4%

Stablecoin bridge has lower risk in **29.4%** of pair-months.

Risk shapes local capital; the rotation extends beyond it.